

Precept 1

Goals

- Review the concepts of sample space, outcome, events, and probability measures.
- Apply the axioms of probability to prove some properties.
- Illustrate how different modeling assumptions result in different probabilities for the same "question".

Exercises

Exercise 1 (Probability Rules)

Let $\mathbb{P}$ be a probability measure defined on $(\Omega, \mathcal{F})$. Prove the following properties:

1. $\mathbb{P}(\emptyset) = 0$
2. $\mathbb{P}(A \cap B^C) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$
3. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

Solutions :

1. Note that $\Omega \cap \emptyset = \emptyset$ and $\emptyset \cap \emptyset = \emptyset$ (hence mutually exclusive), and $\Omega \cup \emptyset \cup \emptyset \cdots = \Omega$. By axiom 2, axiom 3, and axiom 1, we have that

$$1 = \mathbb{P}(\Omega \cup \emptyset \cup \emptyset \cdots) = \mathbb{P}(\Omega) + \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) \cdots \geq \mathbb{P}(\Omega) + \mathbb{P}(\emptyset).$$

So $\mathbb{P}(\emptyset) \leq 1 - 1 = 0$. Using axiom 1 again, we get $\mathbb{P}(\emptyset) = 0$.

2. Note that $A \cap B$ and $A \cap B^C$ are disjoint. Moreover,

$$(A \cap B) \cup (A \cap B^C) = A.$$

By axiom 3, it follows that $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^C)$, and our desired result follows.

3. Note that B and $A \cap B^C$ are mutually exclusive. Also,

$$B \cup (A \cap B^C) = B \cup A.$$

Thus, we can write $\mathbb{P}(A \cup B) = \mathbb{P}(B) + \mathbb{P}(A \cap B^C)$. Applying the previous property, the desired result follows.

Exercise 2 (Assigning Probabilities to Events)

Suppose you have an unfair four-sided die (think triangular pyramid); where an even number has twice the probability of coming up as an odd number. You throw the die once and record the number on the face that's facing down.

1. What is the sample space of this random experiment?
2. Define the probability measure on this sample space.
3. Find the probability that you get a number less than 3.

Solutions:

1. A good sample space for this problem is $\Omega = \{1, 2, 3, 4\}$.
2. We have the following modeling assumption: *we are twice as likely to get an even number than an odd number.*

Based on this modeling assumption, if the probability of getting an odd number is p , then the probability of getting an even number is $2p$. By axiom (ii), we have $\mathbb{P}(\Omega) = 1$. Since the outcomes are mutually exclusive, it follows by axiom (iii) that

$$\begin{aligned}\mathbb{P}(\Omega) = 1 &= \mathbb{P}(\{1\}) + \mathbb{P}(\{2\}) + \mathbb{P}(\{3\}) + \mathbb{P}(\{4\}) \\ &= p + 2p + p + 2p \\ &= 6p \implies p = 1/6.\end{aligned}$$

So $\mathbb{P}(\{1\}) = \mathbb{P}(\{3\}) = 1/6$ and $\mathbb{P}(\{2\}) = \mathbb{P}(\{4\}) = 2/6 = 1/3$.

3. Let A be the event of getting a number less than 3. Clearly, $A = \{1, 2\}$ and $\mathbb{P}(A) = \mathbb{P}(\{1, 2\}) = \mathbb{P}(\{1\}) + \mathbb{P}(\{2\}) = 1/6 + 1/3 = 1/2$.

Exercise 3 (Probability Paradox: The Importance of Precise Modeling)

1. *Cutting a cake:* It's Jessica's birthday and her friends throw her a surprise birthday party. Naturally, they bought her a chocolate cake with raspberry mousse, her favorite. She blows out the candles, and just can't wait to taste it, but unfortunately, her friends are quite absent-minded and they forgot to bring a suitable knife. Everyone goes searching, and for better or worse, that anime-loving friend of theirs brings out their katana sword. In the moment of excitement, and lack of judgment, they decide to play the following game: everyone gets blindfolded and gets to take exactly one slice of the cake. The first person whose cut is larger than the radius of the cake wins and gets to eat the first slice. Naturally, Jessica goes first. What is the probability that she wins?

Mathematically this question essentially asks: *Given a fixed circle with radius r , what is the probability that a randomly selected chord has length greater than or equal to r ?*

You might immediately notice that this problem exhibits the same kind of "imprecision" in this problem. There needs to be a fixed modeling assumption before we can go on with the problem. The core issue is: what does *randomly selected* mean? We need to fix a sample space and a probability measure; there are many reasonable ways that we can do this and we'll explore three possibilities now.

- (a) **Independently and uniformly choose two points from the circumference of the circle to create a chord.**
- (b) **Choosing the midpoint of the chord uniformly from the interior of the circle.**
- (c) **Choosing the perpendicular distance of the chord to the center of the circle uniformly at random.** This one sounds a bit less intuitive. The sequence of steps is 1) choose a radius, 2) choose a random point on the radius (uniformly), and 3) draw the chord perpendicular at that point.

For each of the possibilities, answer the following questions:

- What is the sample space?
- How do we assign probabilities, i.e. what does "chosen uniformly" mean?

Solutions :

- (a) First, let us fix the orientation and choose any fixed starting point A . Now, a second point B is chosen uniformly from the circumference and our job is to calculate the probability that the chord AB has a length greater than r . *What is the sample space?*

$$\Omega = \{\text{points on the circle}\}.$$

How do we assign probabilities, e.g., what does the term "chosen uniformly" mean? Similarly to the homework, we could say that the probability of choosing point B to be in the subset $E \subseteq \Omega$ to be

$$\mathbf{P}(E) = \frac{\text{length}(E)}{\text{circumference of the circle}}.$$

(Draw an example.) In particular, this means that the probability of landing in any given arc is simply the length of the arc divided by the circumference.

Intuitively, we notice that there are more "long chords" than "short chords" (draw a circle of radius r with center A), so instead, let us consider the simpler event:

$$\{AB \text{ is longer than } r\} = \Omega \setminus \{AB \text{ is shorter than } r\}.$$

Now, it is easy to show via an equilateral triangle with two radii and one chord that if a chord cuts an arc of angle $\pi/3$, then the chord has a length equal to the radius (corresponds to the case of an equilateral triangle). If it cuts an arc that is less than $\pi/3$, then the length of the chord is less than r . Let B_1 be a point on the circle $\pi/3$ to the "left" of A and B_2 be a point on the circle $\pi/3$ to the "right" of A . Therefore, if B is on the arc B_1B_2 , then the chord AB has a length less than r . So,

$$\mathbf{P}\{AB \text{ is shorter than } r\} = \mathbf{P}\{B \text{ is on } B_1B_2\} = \frac{\pi/3 + \pi/3}{2\pi} = \frac{1}{3}.$$

Our desired probability is thus $\boxed{\frac{2}{3}}$.

- (b) This is an equally plausible way of modeling the randomness in this problem. To draw a connection to the homework, you might consider throwing a dart at the circle, and then drawing the (unique) chord for which the dart is the midpoint. *What is the sample space?*

$$\Omega = \{\text{points on the interior of the circle}\}.$$

How do we assign probabilities, e.g., what does the term “chosen uniformly” mean? Exactly as suggested by the homework, we could say that the probability of choosing a midpoint M to be in the subset $E \subseteq \Omega$ to be

$$\mathbf{P}(E) = \frac{\text{area}(E)}{\text{area of the circle}}.$$

Let d be the distance between M and the center of the circle. The length of the chord is

$$c = 2\sqrt{r^2 - d^2},$$

so if we want $c \geq r$, we have

$$\frac{\sqrt{3}}{2}r \geq d.$$

What does this mean? (Draw concentric circles.) The inner circle, call it $\tilde{C}$, has radius $\frac{\sqrt{3}}{2}r$ and we note that any point M chosen in this circle has a corresponding chord with length longer than r . We have

$$\mathbf{P}\{\text{chord from } M \text{ has length longer than } r\} = \mathbf{P}\{M \in \tilde{C}\}$$

To compute this probability, we need to compute the ratio of the areas of the two circles:

$$\boxed{\frac{3}{4}}.$$

The goal of this example is the same as the goal of the third problem on the homework; subtle differences in the modeling assumptions can change the problem dramatically.

- (c) First, let us fix the orientation of the circle by choosing a radius AB . *What is the sample space?*

$$\Omega = \{\text{points on the radius } AB\}.$$

How do we assign probabilities, e.g., what does the term “chosen uniformly” mean? Like the first example, we could say that the probability of choosing the point M to be in the subset $E \subseteq \Omega$ to be

$$\mathbf{P}(E) = \frac{\text{length}(E)}{\text{length of } AB}.$$

Let d be the distance from M to the center of the circle. Following the *exact same steps as the previous example*, we can deduce that for the chord to be longer than the radius, it must be that

$$\frac{\sqrt{3}}{2}r \geq d.$$

However, we have defined the probability measure in a very different way. Using the new definition, we have

$$\begin{aligned} \mathbf{P}\{\text{chord from } M \text{ has length longer than } r\} &= \mathbf{P}\left\{d \in \left[0, \frac{\sqrt{3}}{2}r\right]\right\} \\ &= \boxed{\frac{\sqrt{3}}{2}}. \end{aligned}$$

2. Probability Paradox 2: Choose a number

- (a) A real number between 0 and 10 is chosen at random. What is the probability that it is greater than 5?
- (b) A real number between 0 and 100 is chosen at random. What is the probability that it is greater than 25?
- (c) What is the probability that the square of a random number from $[0, 10]$ is greater than 25?
- (d) What is the probability that the square root of a random number from $[0, 100]$ is greater than 5?

Solutions:

- (a) The sample space is $\Omega = [0, 10]$, and the probability will be equal to $\frac{1}{2}$.
- (b) The sample space is $\Omega = [0, 100]$, and the probability will be equal to $\frac{3}{4}$.
- (c) This event is the same as the event of choosing a number that is greater than 5 from $[0, 10]$, and thus the probability is $\frac{1}{2}$.
- (d) This event is equivalent to choosing a number greater than 25 from $[0, 100]$ and thus the probability is $\frac{3}{4}$. However notice that if we look at the interval $[0, 10]$ we are essentially choosing in a random manner a number that is greater than 5, but we have calculated that such choice should yield probability $\frac{1}{2}$. Is this a contradiction? A paradox? Well, the thing is that we are dealing with different sample spaces here and different probability measures. Taking a square or a square root respectively skews the uniform nature of the probabilities assigned and thereby yields a different answer. Thus to avoid such "paradoxes" one has to specify both the sample space and the probability measure exactly.